P1

a) $\phi(\rho,\varphi) = a_0 + b_0 \ln\rho + \sum_{k=1}^{\infty}(\alpha_k \rho^k + \beta_k \rho^{-k})(a_k \cos k\varphi + b_k \sin k\varphi)$

b) $\sigma(\varphi) = \sigma_0 \sin 5\varphi$

Inside and outside the cylinder $\nabla^2\phi = 0$

$\Rightarrow$ use (a) and match BC so that (1) $\phi(R-0,\varphi) = \phi(R+0,\varphi)$

(2) $E_n(R+0,\varphi) - E_n(R-0,\varphi) = \frac{\sigma(\varphi)}{\varepsilon_0} = \frac{\sigma_0}{\varepsilon_0}\sin 5\varphi.$

$$\vec{E} = -\vec{\nabla}\phi = -\left(\frac{\partial\phi}{\partial\rho}\hat{\rho} + \frac{1}{\rho}\frac{\partial\phi}{\partial\varphi}\hat{\varphi} + \frac{\partial\varphi}{\partial z}\hat{z}\right)$$

$\Rightarrow E_n = -\frac{\partial\phi}{\partial\rho}$ here.

$0<\rho<R$: $\beta_k = 0$, $b_0 = 0$.

$R<\rho<\infty$: $\tilde{\alpha}_k = 0$, $\tilde{a}_0 = 0$

where I use that $\phi(\rho>R,\varphi) = \tilde{a}_0 + \tilde{b}_0 \ln\rho + \sum_{k=1}^{\infty}(\tilde{\alpha}_k \rho^k + \tilde{\beta}_k \rho^{-k}) \cdot$

$\cdot(\tilde{a}_k \cos k\varphi + \tilde{b}_k \sin k\varphi)$

BC (1) at $\rho=R$ tells us that: $a_0 = 0$, $\tilde{b}_0 = 0$

Surface charge density $\sigma(\varphi)$ is the source of electric field

$\Rightarrow$ all harmonics except $k=5$ are absent.

$\Rightarrow$ $\phi(\rho<R,\varphi) = \alpha_5 b_5 \sin(5\varphi)\,\rho^5$

$\phi(\rho>R,\varphi) = \tilde{\beta}_5 \tilde{b}_5 \sin(5\varphi)\,\rho^{-5}$ $\Rightarrow$ $\tilde{\beta}_5\tilde{b}_5 = R^{10}\alpha_5 b_5.$

BC (2): $E_n(\rho=R+0) = +\frac{5\tilde{\beta}_5\tilde{b}_5\sin(5\varphi)}{R^6}$

$E_n(\rho=R-0) = -5R^4\alpha_5 b_5 \sin(5\varphi)$

$\Rightarrow \frac{5\tilde{\beta}_5\tilde{b}_5}{R^6} - (-5R^4\alpha_5 b_5) = \frac{\sigma_0}{\varepsilon_0} \Rightarrow 10R^4\alpha_5\beta_5 = \frac{\sigma_0}{\varepsilon_0}.$

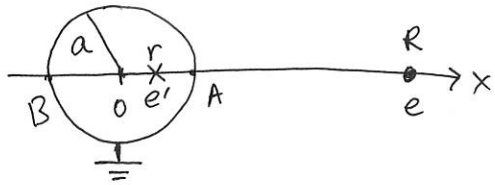
$\Rightarrow \tilde{\beta}_5\tilde{b}_5 = \frac{\sigma_0}{10\varepsilon_0}R^6$

$$\Rightarrow \Phi(\rho < R, \varphi) = \frac{\sigma_0}{10\,\varepsilon_0} \frac{\rho^5}{R^4} \sin(5\varphi)$$

$$\Phi(\rho > R, \varphi) = \frac{\sigma_0}{10\,\varepsilon_0} \frac{R^6}{\rho^5} \sin(5\varphi)$$

P2

Grounded sphere
$\Rightarrow \phi(r=a)=0.$



Try image charge e' at distance $r<a$ on x axis.
e' is such that ~~[illegible]~~ potential of the sphere is 0.

At point A: $\phi_A = \dfrac{e}{4\pi\varepsilon_0(R-a)} + \dfrac{e'}{4\pi\varepsilon_0(a-r)} = 0$

@ B: $\phi_B = \dfrac{e}{4\pi\varepsilon_0(R+a)} + \dfrac{e'}{4\pi\varepsilon_0(a+r)} = 0$

$$\Rightarrow \begin{cases} e(a-r)+e'(R-a)=0 \\ e(a+r)+e'(R+a)=0 \end{cases} \Rightarrow \begin{cases} ea+e'R=0 \\ er+e'a=0 \end{cases}$$

$$\Rightarrow \quad e' = -\frac{ea}{R}, \quad r = -\frac{e'a}{e} = +\frac{a^2}{R}.$$

Force on e: $F = \dfrac{ee'}{4\pi\varepsilon_0(R-r)^2} = -\dfrac{e^2 a}{4\pi\varepsilon_0 R}\dfrac{1}{\left(R-\frac{a^2}{R}\right)^2} = -\dfrac{e^2 aR}{4\pi\varepsilon_0(R^2-a^2)^2}$

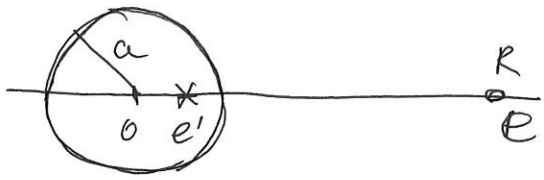
for $R \gg a$: $F = -\dfrac{e^2 a}{4\pi\varepsilon_0 R^3}$

for $R=a+\delta$: $F \approx -\dfrac{e^2 a(a+\delta)}{4\pi\varepsilon_0\,\delta^2(2a+\delta)^2} \underset{\delta \ll a}{=} \dfrac{-e^2a^2}{4\pi\varepsilon_0\cdot 4a^2\delta^2} =$

$$=(\delta = R-a) = -\frac{e^2}{16\pi\varepsilon_0(R-a)^2}$$

This is the same as the force between e and the conducting plane because for $R \approx a$ (e very close to the sphere) the curvature of the sphere is not noticable.

(P3) Isolated sphere $\Rightarrow Q_{tot} = 0$.



With one image charge $e' = -\frac{ea}{R}$ @ $r = \frac{a^2}{R}$
potential of the sphere $= 0$ but its charge is $e' \neq 0$.
For an isolated sphere its total charge is 0
but the sphere remains to be equipotential (even though $\phi_{sphere} \neq 0$ anymore)
$\Rightarrow$ to achive this add another $e'' = -e'$ at the origin $r = 0$.

$$\Rightarrow Q_{tot} = e' + e'' = 0$$

a) $$\phi_{sphere} = \underbrace{\phi_e + \phi_{e'}}_{0} + \phi_{e''} = \frac{e''}{4\pi\varepsilon_0 a} = \frac{e}{4\pi\varepsilon_0 R}.$$

b) Force on the charge e:

$$F = \frac{ee'}{4\pi\varepsilon_0 (R-r)^2} + \frac{ee''}{4\pi\varepsilon_0 R^2} = -\frac{e^2 a}{4\pi\varepsilon_0 R}\frac{1}{(R-r)^2}$$

$$+ \frac{e^2 a}{4\pi\varepsilon_0 R}\frac{1}{R^2} = \frac{e^2 a}{4\pi\varepsilon_0}\left(\frac{1}{R^3} - \frac{1}{R(R-a^2/R)^2}\right) =$$

$$= \frac{e^2 a}{4\pi\varepsilon_0 R}\left(\frac{1}{R^2} - \frac{R^2}{(R^2-a^2)^2}\right) = \frac{e^2 a}{4\pi\varepsilon_0 R}\left(\frac{1}{R^2} - \frac{R^2}{R^4\left(1-\frac{a^2}{R^2}\right)^2}\right)$$

$$= \frac{e^2 a}{4\pi\varepsilon_0 R^3}\left(1 - \left(1 + \frac{2a^2}{R^2}\right)\right) = -\frac{e^2 a^3}{2\pi\varepsilon_0 R^5}.$$

This is much smaller than $-\frac{e^2 a}{4\pi\varepsilon_0 R^3}$ found in P2. for $R \gg a$.